Course Project: Item-Level Price Co-Movement Network using CPI Data

Discrete Structures

1. Project Objective

The objective of this project is to model and analyze the similarity of consumer items based on their price movement patterns across multiple cities. Instead of representing cities as graph nodes, this project represents **items (products) as nodes**. Two items are considered adjacent if a large number of cities exhibit similar price increase/decrease patterns for the two items.

Students will apply concepts of relations, graphs, similarity measures, and centrality metrics to construct and analyze an item–item network. The project also includes a temporal component using three years of monthly CPI data.

2. Data Description

The CPI data is provided on a monthly basis for multiple cities and multiple consumer items.

The dataset spans **three consecutive years**. Each record contains the price of an item in a specific city and month.

Students must use the following dataset:

`https://www.pbs.gov.pk/price/`

3. Node and Feature Representation

Let:

- I be the set of items,
- C be the set of cities,
- T_y be the set of months of year y.

Each item is treated as a node in the network.

For a fixed city c and item i, students must compute a **price-change pattern vector** for each year y:

$$\mathbf{v}_{i,c}^{(y)} = (\Delta p_{i,c}^{(y,2)}, \Delta p_{i,c}^{(y,3)}, \ldots, \Delta p_{i,c}^{(y,12)}),$$

where $\Delta p_{i,c}^{(y,m)}$ denotes the price change from month $m-1$ to month m.

4. Item Similarity within a City

For two items i and j in a city c and year y, students must compute the cosine similarity:

$$\mathrm{sim}_c^{(y)}(i,j) = \frac{\mathbf{v}_{i,c}^{(y)} \cdot \mathbf{v}_{j,c}^{(y)}}{\|\mathbf{v}_{i,c}^{(y)}\| \, \|\mathbf{v}_{j,c}^{(y)}\|}.$$

This measures how similar the monthly price-change patterns of the two items are within that city for that year.

5. Aggregation across Cities

For a fixed year y, define

$$N_y(i,j) = \left| \left\{ c \in C \mid \mathrm{sim}_c^{(y)}(i,j) \geq \tau \right\} \right|,$$

where τ is a similarity threshold.

6. Graph Construction for a Single Year

For each year y, construct an undirected graph

$$G_y = (I, E_y),$$

where

$$(i,j) \in E_y \iff N_y(i,j) \geq K,$$

and K is a user-defined city-count threshold.

Hence, two items are connected if a sufficiently large number of cities show similar price movement patterns for those items in that year.

7. Weighted Graph Variant

Students must also construct a weighted version of the network where

$$w_y(i,j) = N_y(i,j),$$

or alternatively

$$w_y(i,j) = \frac{1}{|C|} \sum_{c \in C} \mathrm{sim}_c^{(y)}(i,j).$$

8. Year-wise Pattern Analysis

Students must construct and analyze a separate graph for each of the three years:

$$G_{y_1}, \; G_{y_2}, \; G_{y_3}.$$

For each yearly graph, students must compute and interpret:

- degree centrality,
- closeness centrality,
- betweenness centrality.

Students must identify highly central items and interpret their role in terms of price co-movement.

9. Temporal Network Analysis across Three Years

Let the three yearly graphs be ordered in time.

Students must study:

- appearance and disappearance of edges across years,
- stability of highly central items across years,
- changes in connected components or clusters of items.

A temporal comparison must be performed to identify persistent and transient item relationships.

10. Category-based Extension

Each item belongs to one of the predefined seven categories. Students must analyze:

- connectivity within categories,
- connectivity between categories,
- whether clusters of items align with category boundaries.

11. Threshold and Weight Sensitivity Analysis

Students must analyze the effect of:

- varying the similarity threshold τ,
- varying the city-count threshold K,
- varying the weighting schemes for edges.

The resulting changes in graph structure and centrality rankings must be clearly reported.

12. Required Implementation

Students must implement the complete pipeline:

1. data preprocessing,
2. construction of yearly item similarity matrices,
3. construction of yearly graphs,
4. computation of centrality measures,
5. temporal comparison across three years.

The implementation must be reproducible.

13. Demo Video Requirement

Students must produce a demonstration video that includes:

- visualization of at least two different weighting schemes,
- visual comparison of graphs for different thresholds,
- discussion of the effect of cosine similarity thresholds,
- discussion of changes in centrality rankings under different weights,
- explanation of temporal changes across the three yearly networks.

The video must be uploaded to YouTube and the link must be provided in the report.

14. Deliverables

Each group must submit:

- a technical report describing methodology and results,
- source code used for the analysis,
- a YouTube link to the demonstration video.

15. Learning Outcomes

After completing this project, students should be able to:

- model real-world data using relations and graphs,
- construct item-based similarity networks,
- apply cosine similarity for temporal pattern comparison,
- analyze graphs using standard centrality measures,
- reason about temporal evolution of networks.